

1. (a) Discuss how the creation and use of computer viruses is prohibited under the Computer Misuse Act.

[3]

(b) The General Data Protection Regulation (GDPR) was adopted on 14 April 2016 and became enforceable beginning 25 May 2018.

Discuss four reasons why potential societal issues introduced by new technologies may require GDPR be amended, or new legislation be introduced.

[4]

(c) A company has applied to copyright a database they have created. Their database contains a simple ordering of 10,000 novels alphabetically by title.

Discuss whether the company is likely to succeed in their attempt to copyright the database described above.

[4]

(d) A researcher is planning to conduct an experiment to investigate if a chatbot they have developed helps novice Java programmers solve lab exercises during a programming course.

The researcher plans to conduct the experiment on a Java Programming course they will teach next year during the mandatory tutorial sessions within the course.

Identify two potential ethical concerns with this research.

[4]

(e) You have been tasked with creating a survey to investigate consumers' frustration using a new virtual reality application, "MBP VR".

Describe one closed and one open-ended question that you could use in the survey to capture this.

[2]

(f) Describe two changes you might make to your closed question if you were conducting a study with child participants (e.g., aged 6-10).

[2]

2. Consider the following experimental design and evaluation methods:

A researcher has designed an experiment to evaluate the impact of latency on the usability of keyboards.

The conditions are:

- Condition 1: A keyboard with 5 milliseconds added latency
- Condition 2: A keyboard with 25 milliseconds added latency
- Condition 3: A keyboard with 50 milliseconds added latency

During the experiment, participants experience each of the three conditions sequentially. All experience Condition 1 first, then Condition 2, then Condition 3.

For each condition, participants are tasked to type out the same set of sentences in the same order.

The researcher records the typing speed and number of errors made by participants for each condition.

The researcher plans to calculate the total number of errors made for each condition and conclude that the condition with the lowest number of errors has the best performance.

(a) Identify the independent and dependent variables in the proposed experiment design. [3]

(b) Critique the experiment design and analysis methodology proposed by the researcher to investigate the stated research goal by explaining negative aspects of the presented approach.

Propose changes to address any identified negative issues. [15]

(c) Describe two reasons why inductive inference would be used to evaluate the hypothesis of this experiment opposed to deductive inference. [2]

3. Locomotion systems in VR games provide ways for a player to navigate around a virtual environment presented by the VR system.

You have designed a new locomotion technique for a VR game you have made.

You believe your new locomotion system (“LocoPro”) will provide a better player experience by being easier to control and more immersive experience than the current locomotion system (“LocoOriginal”).

You plan to conduct a research project with an experiment (with human participants) to investigate this belief.

(a) Propose a hypothesis for an experiment to investigate this belief.

[2]

(b) Propose an experiment to investigate this hypothesis.

You should cover all aspects of a research project, except performing a literature survey and writing a research proposal.

[19]